SWALLOHear: A Reliability-Aware Acoustic Swallowing Analysis Research System for Post-Stroke Dysphagia

Research paper and system documentation

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> **Research prototype, not a diagnostic device.** SWALLOHear detects swallowing-related sounds in neck-microphone audio and reports how far each result can be trusted. It does not diagnose dysphagia, aspiration or pneumonia, does not grade severity, and must not be used to make care decisions.

Abstract

Background. About 42% of people with acute stroke develop dysphagia (difficulty swallowing), which is associated with roughly four-fold higher odds of pneumonia and death (Banda et al., 2022). Instrumental swallowing tests need X-ray equipment and specialists. Cervical auscultation, recording swallowing sounds at the neck, is cheap and non-invasive, but automatic analysis of these sounds is rarely evaluated on unseen patients, on noisy recordings, or for whether it knows when it is wrong.

Methods. We built SWALLOHear, a reliability-aware pipeline for neck-microphone audio, using the public *Swallowing High Resolution Cervical Auscultation Signals* dataset (Khalifa et al., 2021). After auditing the files (1,090 usable labelled recordings from 174 participants) we made a participant-disjoint 70/15/15 split, so no person appears in more than one part. We compared an energy-threshold baseline with a Random Forest on 81 acoustic features, trained a noise-augmented Random Forest, and added a reliability layer: a signal-quality gate that withholds results on poor recordings, and a separately trained reliability model that scores each detected event. Every threshold was chosen on validation participants; the 27 test participants (171 recordings, 288 swallows) were used once. Confidence intervals come from a participant-level bootstrap.

Results. On clean audio the Random Forest reached event F1 0.50 (95% CI 0.40–0.59) against 0.28 (0.23–0.34) for the energy baseline. A clean-trained model collapsed under white noise (F1 0.14 at 20 dB, 0.00 at 10 dB), while the noise-augmented model kept 0.45, 0.39 and 0.31 at 20, 10 and 5 dB. The detector's own confidence barely separated correct from wrong detections (AUROC 0.53, 0.46–0.60); the learned reliability

score did (0.79, 0.73–0.83). Reporting only events the reliability model calls *Confident* raised precision from 0.41 to 0.69 and cut false detections from 9.4 to 1.5 per minute, at the cost of recall (0.63 → 0.33). The quality gate withheld 2% of clean recordings and 82% and 94% at 10 and 5 dB. Temperature scaling did not improve calibration. A 30-second clip is analysed in 0.13 s on a laptop CPU.

Conclusions. Absolute accuracy is modest, but SWALLOHear makes its failures visible: it says when a recording is too poor to judge and which detections to trust. It is delivered as an open repository, a web application and a mobile application for Android and iOS, all running the same code.

Keywords: cervical auscultation; swallowing-event detection; post-stroke dysphagia; reliability; selective prediction; signal quality; participant-independent validation; reproducible research.

How to read this document

If you are…	Read
A judge or reader who wants the research	Abstract, sections 1, 3, 4, 5, 8
A clinician, therapist or student who wants to **use** the app	Section 2 (plain-language overview), then section 7 (user guide)
A developer who wants to run or extend it	Sections 6 and 7, and the appendices

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Contents

1. Introduction
2. SWALLOHear in plain language
3. Data
4. Methods
5. Results
6. The SWALLOHear system
7. User guide
8. Discussion
9. Future work
10. Conclusion
11. References
12. Appendices

1. Introduction

1.1 Stroke and dysphagia

Difficulty swallowing is one of the most common complications of stroke. A meta-analysis of 42 studies with 26,366 patients found a pooled dysphagia prevalence of 42% in acute stroke. Affected patients had markedly higher odds of pneumonia (odds ratio 4.08, 95% CI 2.13–7.79) and death (4.07, 2.17–7.63) (Banda et al., 2022). How often dysphagia is found depends on how carefully it is looked for: 37–45% with screening, 51–55% with clinical testing and 64–78% with instrumental testing (Martino et al., 2005). In the Philippines, reported stroke prevalence ranges from 0.5% to 6.0% across studies (Collantes et al., 2022). Early, objective and affordable assessment of swallowing is therefore a public-health research priority.

1.2 Cervical auscultation

The reference test, videofluoroscopy, films a patient swallowing under X-ray. It gives exact timing but needs equipment, radiation and trained staff. Bedside screening is quick but coarse. **Cervical auscultation** records the sounds and vibrations of swallowing at the neck with a microphone or accelerometer. The effect of dysphagia on these signals has been reviewed (Dudik et al., 2018), and the hardware is cheap, which makes it an attractive basis for affordable research tools, provided the analysis can be trusted.

Prior work on the same public data trained a deep neural network on spectrogram windows and reported per-record sensitivity of 85.3% and specificity of 83.8% (Khalifa et al., 2020). That is strong work, with two features that shape ours: its best channel was an accelerometer, not the microphone that a phone or stethoscope provides; and window-level accuracy does not measure event start/end timing or tell a user when not to trust a detection.

1.3 The problem

Detecting that a swallow happened is not the same as measuring it reliably, and neither is the same as knowing when not to trust the measurement. Three weaknesses limit current practice:

1. **Optimistic evaluation.** If recordings of the same person appear in both training and test data, results look better than they will on new patients (Saeb et al., 2017).
2. **Window-level reporting.** Accuracy on short windows hides errors in event start, end and duration, which feed into any downstream measure such as swallowing rate.
3. **Silent failure.** A noisy, clipped or low-quality recording can produce confident-looking but wrong output, and the user cannot tell. Modern models are often over-confident (Guo et al., 2017). This risk is greatest exactly where cheap tools are most needed: noisy bedside and low-resource settings.

1.4 Research questions and hypotheses

Central question. Under participant-disjoint evaluation, how accurately can a reliability-aware acoustic pipeline find swallowing events in neck-microphone audio, and can it flag the detections and recordings that should not be trusted?

- **RQ1 (localisation).** How do an energy baseline and machine-learning detectors compare in event-level F1 and start/end error on unseen participants?
- **RQ2 (reliability).** Do calibration, a learned reliability score and a signal-quality gate raise the precision of reported events and expose failures under noise, and what coverage is sacrificed?
- **RQ3 (accessibility).** How does performance degrade under noise, and can the whole workflow run interactively on a CPU, in a browser and on a phone?

Pre-specified hypotheses (from our proposal): **H1** a compact CNN beats the best conventional detector in event F1; **H2** at 5 dB SNR, events labelled Confident have higher precision than unfiltered detections; **H3** temperature scaling lowers expected calibration error. Their outcomes, including the negative ones, are in section 5.8.

1.5 Contributions

1. A **participant-disjoint, event-level evaluation** of microphone-only swallowing detection, with participant-level bootstrap confidence intervals and a self-checking reproduction script.
2. A **reliability layer**: a data-driven signal-quality gate plus a learned event-reliability model that is far better than the detector's own confidence at telling right from wrong, with an exact per-event explanation.
3. Evidence that **noise-augmented training** prevents the collapse of a clean-trained detector under noise.
4. An **open, usable tool**: a web application, a mobile application for Android and iOS, and an HTTP API, all running the same code as the experiments.

1.6 Scope

SWALLOHear does	SWALLOHear does not
Detect and time swallowing-related acoustic events in compatible recordings	Diagnose dysphagia, aspiration or pneumonia
Report signal quality and withhold results it cannot trust	Grade severity or predict outcomes
Score and explain how trustworthy each detection is	Claim clinical or Philippine validation
Measure swallowing rhythm and agreement with expert labels	Replace videofluoroscopy or a clinician

2. SWALLOHear in plain language

This section is for readers without a technical background.

2.1 What it does, in one minute

You give SWALLOHear a sound recording made with a microphone resting against the side of the throat. It:

1. **Checks whether the recording is good enough.** If it is too noisy, distorted, too short or too long, SWALLOHear says **"Result withheld"** and explains why, instead of guessing.
2. **Finds the swallowing sounds.** It marks where each one starts and ends, like highlighting words in a sentence.
3. **Says how much to trust each one.** Every detected sound gets a **reliability score** from 0 to 1. Scores of 0.50 or more are **Confident**; lower scores are **Low-confidence** ("check this one yourself").
4. **Explains itself.** For any detection it shows which measurements pushed the score up or down.
5. **Summarises.** It counts events, measures the rhythm of swallowing, compares with an expert's labels when they exist, and lets you download everything or print a report.

2.2 The four possible verdicts

Verdict	What it means	What to do
✓ **Events detected**	The recording was good enough, and swallowing-related sounds were found.	Look at each event's tier (Confident or Low-confidence).
○ **No event detected**	The recording was good enough, but nothing looked like a swallow.	This does not prove no swallow happened.
! **Result withheld** (Inconclusive)	The recording was too poor to judge. The reason is shown (e.g. *Low SNR*: too noisy).	Record again in a quieter place, or adjust the microphone.

Within "Events detected", each event is either **Confident** (teal) or **Low-confidence** (amber).

2.3 Key words

Word	Meaning
Event	One detected swallowing-related sound, with a start time, end time and duration.
SNR (signal-to-noise ratio)	How much louder the swallowing sounds are than the background noise, in decibels (dB). Higher is better; the app needs about 11.5 dB.
Reliability score	A number from 0 to 1 saying how likely an event is to be a real, correctly placed swallow.
Expert labels	The swallow times marked by specialists from X-ray video, available for the demo clips.
Precision	Of the events the app reports, the share that are real.
Recall	Of the real swallows, the share the app finds.
F1	One number combining precision and recall (1.0 is perfect, 0 is useless).

2.4 Why "knowing when not to trust" matters

A tool that is right half the time but cannot tell which half is hard to use. SWALLOHear's main idea is to separate the detections it can stand behind from those it cannot, and to refuse to answer on recordings it cannot read. When it says *Confident*, it is right about 7 times in 10 on clean test recordings; when it reports everything, it is right about 4 times in 10 (section 5.3).

3. Data

3.1 Source

We used **Swallowing High Resolution Cervical Auscultation Signals** (Khalifa, Donohue, Coyle & Sejdić, 2021; University of Pittsburgh; Zenodo, DOI 10.5281/zenodo.4539695; licence CC BY 4.0). It contains swallowing recordings from patients referred for videofluoroscopy, with swallow start and end times marked on the synchronised X-ray video. No new patient data were collected for this study.

3.2 What the files contain

Item	Detail
Signals	`signals/<id>.npy`: an *N* × 4 array sampled at 4 kHz. Channel 0 is the **microphone** (our input); channels 1–3 are a three-axis accelerometer (anterior–posterior, superior–inferior, medial–lateral).
Labels	`labels/<id>.csv`: one row per swallow with start and end **video frame numbers at 60 frames/s**.
Time conversion	sample = frame × 4000 / 60, so label resolution is about **17 ms**. Timing differences smaller than that are not meaningful.
Recording names	e.g. `ns011a_1`. The prefix `ns` or `s` marks two participant groups whose meaning (for example stroke) is **not documented**; we never treat them as stroke labels.

3.3 Data audit

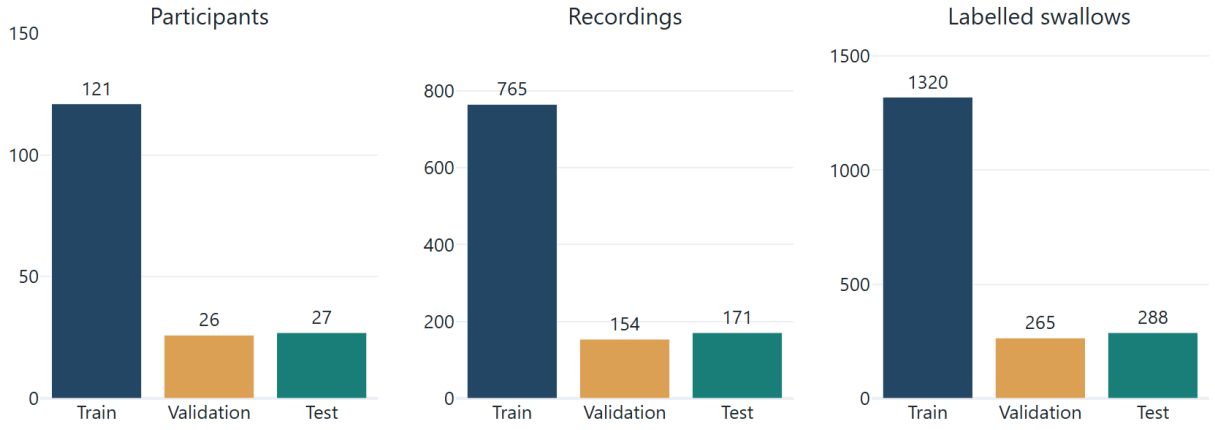
The record page describes 248 patients and 3,144 swallows. The files we downloaded (MD5 checksums verified) contain fewer, and we report what the files contain:

Quantity	Count
Signal files	1,285 recordings
Recordings with a label file	1,092 (174 participants, 1,876 labelled swallows)
Recordings passing sanity checks (labels inside the signal, start < end)	**1,090** (2 excluded)
Recordings without labels	193, used only as extra negative examples in training
Participant groups (`ns` / `s`)	139 / 35 participants
Recording length	median 7.8 s (interquartile range 5.1–11.9 s), longest 45.6 s

3.4 Participant-disjoint split

All recordings of one person stay in one part: **training** (fit the models), **validation** (choose every threshold and setting) or **test** (sealed, used once for the reported numbers). The split is seeded and reproducible (`scripts/build_index.py`), and a unit test checks that the parts share no participant.

Figure 2. Participant-disjoint 70/15/15 split (no person appears in two parts)



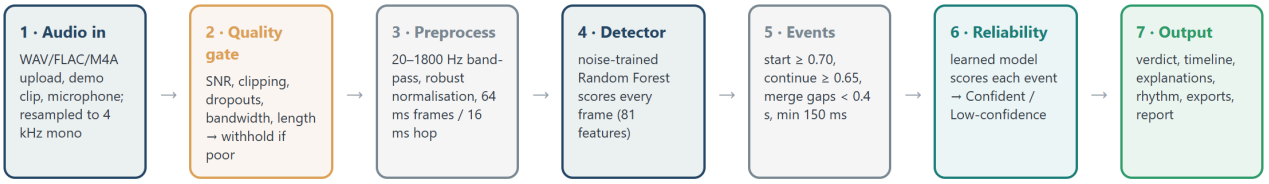
3.5 Demo clips

Seven short clips from **test** participants ship with the apps in `data/demo/` (CC BY 4.0, with attribution), each with its expert labels. One (`ns027d_1_noisy5dB`) has white noise added at 5 dB to show the quality gate.

4. Methods

4.1 Overview

Figure 1. The SWALLOHear pipeline



The web app, the mobile app (through the API) and the experiment scripts all call the same Python package, so what users see is exactly what the experiments measured.

The same Python package (`src/swallohear`) is used by the experiment scripts, the web app and the mobile app's API, so what users see is exactly what was evaluated.

4.2 Audio loading and preprocessing

Step	Setting
Loading	WAV or FLAC (web); also M4A, AAC, MP3, WebM through the API. Stereo is averaged to mono.
Resampling	Any rate → 4 kHz with an anti-aliasing polyphase filter; the original rate is kept to check bandwidth.
Band-pass filter	4th-order Butterworth, 20–1,800 Hz, zero-phase, to remove drift and very high frequencies.
Normalisation	Robust per-recording scaling (median and median absolute deviation), so microphone gain does not dominate.
Framing	64 ms frames (256 samples), 16 ms hop.

4.3 Signal-quality gate

Before trusting any detection, each recording is checked. If any check fails the verdict is **Inconclusive** and the candidate events are shown only for inspection.

Check	Limit	Reason code
Estimated SNR = 95th minus 10th percentile of frame energy	≥ 11.5 dB	`LOW_SNR`
Clipped samples (at the maximum amplitude)	≤ 2%	`CLIPPING`
Flat samples (signal not changing)	≤ 30%	`FLATLINE`
Original bandwidth (half the original sample rate)	≥ 1,000 Hz	`LOW_BANDWIDTH`
Duration	1–120 s	`TOO_SHORT` / `TOO_LONG`

The SNR limit is data-driven: it is the **5th percentile of clean validation recordings**, roughly the worst clean recording the system saw while being tuned.

4.4 Acoustic features

For every frame the Random Forest receives **81 features**: 27 base measurements, plus their mean and standard deviation over ±5 neighbouring frames (176 ms of context).

Base features (27)	
Loudness and texture	log RMS energy, zero-crossing rate
Spectral shape	centroid, bandwidth, 85% roll-off, flatness
Band energies	log energy in 8 bands: 20–100, 100–200, 200–350, 350–500, 500–800, 800–1,200, 1,200–1,600, 1,600–2,000 Hz
Timbre	13 mel-frequency cepstral coefficients (20 mel bands, 20–1,900 Hz)

4.5 Detectors

Detector	**Description**
M0 energy baseline	Short-time energy in 100–1,000 Hz, smoothed over 9 frames, compared with an adaptive threshold (noise floor + 3 × median absolute deviation).
M1 Random Forest (clean-trained)	100 trees, class-balanced, minimum 50 samples per leaf, on the 81 features. Trained on training participants; 25% of non-swallow frames kept to balance classes; unlabelled recordings add negatives.
M1 Random Forest (noise-augmented), the final detector	Same model, but each training recording is used clean **and** with white noise at a randomly chosen 20, 10 or 5 dB.

A compact CNN (M2) was planned in the proposal but **not built** in the time available; the noise-augmented Random Forest is the final detector.

4.6 From frame scores to events

All detectors share the same post-processing (*hysteresis*): an event **starts** when the frame score reaches the start threshold, **continues** while it stays above the continue threshold, events closer than the merge gap are joined, and events shorter than 150 ms are dropped. Values chosen on validation:

Setting	**M0**	**Final Random Forest**
Start threshold	0.30	**0.70**
Continue threshold	0.10	**0.65**
Merge gap	0.4 s	**0.4 s**
Minimum duration	150 ms	**150 ms**

4.7 The reliability layer

First attempt (did not work well). We first followed the proposal: temperature-scale the detector's frame scores, use the mean calibrated score of an event as its confidence, and choose a threshold on validation. Calibration barely changed (section 5.3.4), and the best threshold kept only 16% of real swallows at precision 0.44: raw confidence simply does not know when it is wrong.

Final approach: a learned reliability model. We trained a second model whose only job is to predict whether a detected event is correct (overlaps an expert label by at least half). For each event it sees seven measurements:

#	Measurement	Meaning
1	Mean frame score	Average detector score inside the event
2	Peak frame score	Highest detector score inside the event
3	Score spread	Standard deviation of the score inside the event
4	Duration	Event length in seconds
5	Mean loudness above floor	Average frame energy minus the recording's noise floor (dB)
6	Peak loudness above floor	Loudest frame minus the noise floor (dB)
7	Recording SNR	Estimated signal-to-noise ratio of the whole recording (dB)

The model is a **standardised logistic regression**, trained on events from **training participants only**, using 3-fold participant-grouped *out-of-fold* predictions (so it learns from detections the detector made on people it had not been trained on), on clean audio and at 20, 10 and 5 dB. The **Confident** threshold τ was chosen on validation by a pre-set rule: the highest τ in {0.2, …, 0.7} that raises precision by at least 0.10 over ordinary output while keeping at least 25% of detections. This gave $\boldsymbol{\tau = 0.50}$.

4.8 Explaining a score

Because the reliability model is a standardised logistic regression, every score is an exact sum:

$$\text{log-odds} = \text{intercept} + \sum_i \text{weight}_i \times (\text{value}_i - \text{average}_i) / \text{spread}_i \qquad \text{reliability} = 1 / (1 + e^{(-\text{log-odds})})$$

The intercept is the score of an *average* event; each term is the push from one measurement. The app's **"Why this score?"** chart draws exactly these terms (a waterfall), and they add up to the reliability score shown. We verified this numerically for every event of three demo clips.

4.9 Swallowing rhythm

From the detected events the app reports the event rate per minute, the intervals between consecutive onsets, their mean, the **regularity** (coefficient of variation of the intervals: < 0.25 steady, 0.25–0.5 moderate, > 0.5 irregular) and the longest quiet gap, including the start and end of the recording. Rhythm is descriptive only.

4.10 Evaluation protocol

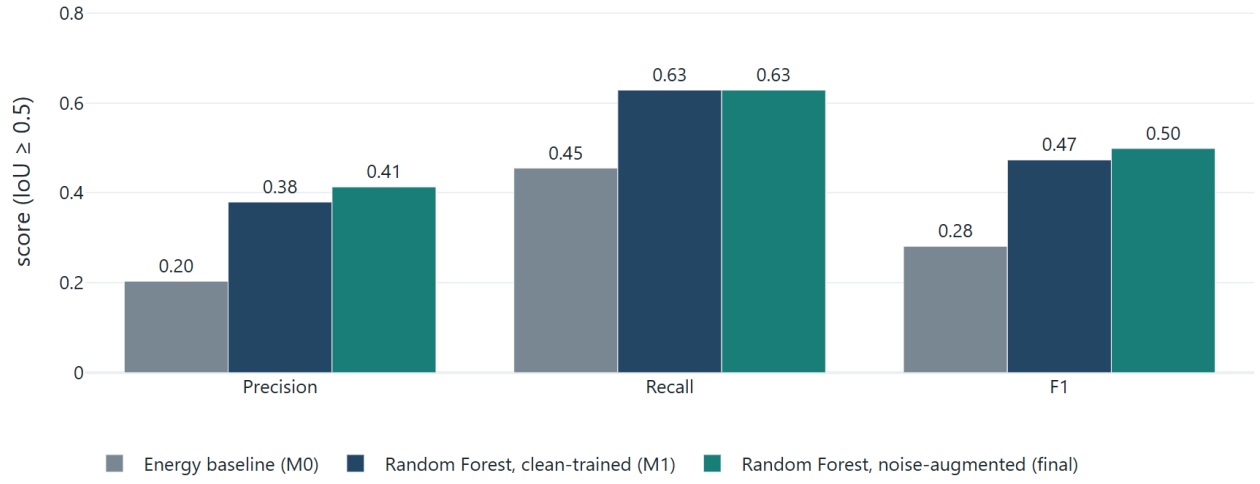
- **Event matching.** A detection is a true positive if it overlaps an unmatched expert label with intersection-over-union (IoU) ≥ 0.5, matched one-to-one (Mesaros et al., 2016). IoU ≥ 0.3 is also reported for the detectors.
- **Metrics.** Precision, recall, F1, false detections per minute, signed and absolute onset/offset error, AUROC for "is this detection correct", expected calibration error (ECE, 15 equal-mass bins).
- **Noise.** White noise added to test recordings at 20, 10, 5 and 0 dB SNR relative to the power inside labelled swallows (fixed seeds).
- **Uncertainty.** 95% confidence intervals from a participant-level cluster bootstrap (resampling the 27 test participants), with `scripts/bootstrap_ci.py`, which also re-computes the main F1 and checks it matches the main run (all four noise levels: "OK").
- **One look at the test set.** All thresholds, τ and the SNR limit were fixed on validation before testing.

5. Results

All results are on the **27 held-out test participants (171 recordings, 288 labelled swallows)** and come from `results/*.json`.

5.1 Detecting swallows on clean audio (RQ1)

Figure 3. Event detection on clean audio, 27 held-out test participants



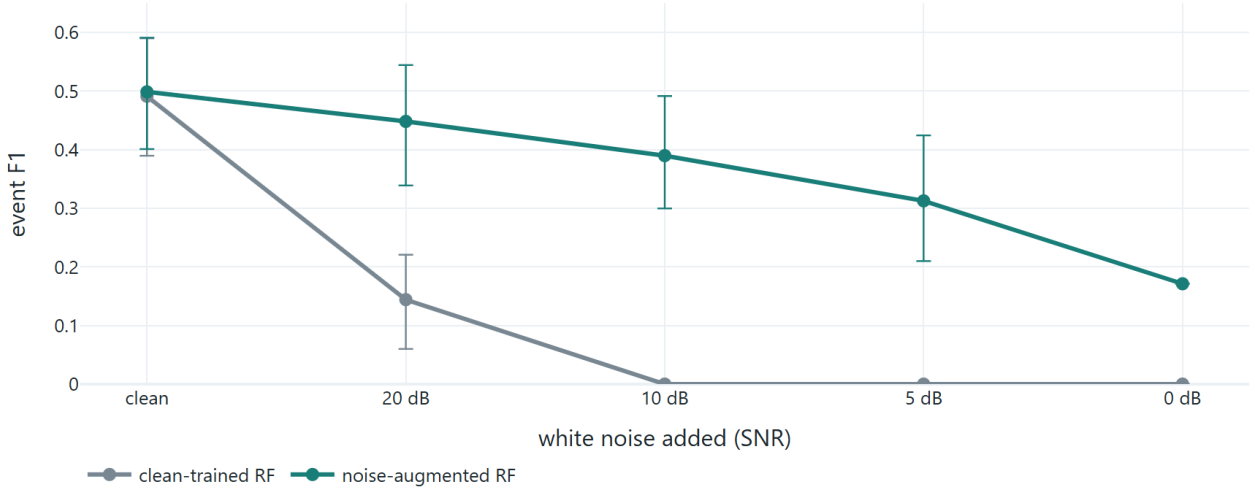
Detector (IoU ≥ 0.5)	Precision	Recall	F1 (95% CI)	False detections / min
M0 energy baseline	0.20	0.45	0.28 (0.23–0.34)	18.8
M1 Random Forest, clean-trained	0.40	0.63	0.49 (0.39–0.59)	9.7
M1 Random Forest, noise-augmented (final)	**0.41**	**0.63**	**0.50 (0.40–0.59)**	**9.4**

With the looser IoU ≥ 0.3 match (baseline script, `results/baselines.json`, which uses a continue threshold of 0.60), F1 is 0.45 for M0 and 0.59 for the Random Forest. On clean audio, noise-augmented and clean-trained Random Forests are tied (difference +0.007, 95% CI −0.012 to +0.027): augmentation costs nothing on clean audio.

Timing (same baseline script). For matched events, the Random Forest's median start error was −57 ms (it fires slightly early) and median end error −22 ms; median absolute errors were 113 ms (start) and 122 ms (end). The energy baseline's were 121 ms and 202 ms. Remember that labels themselves have about 17 ms resolution.

5.2 Robustness to noise (RQ3)

Figure 4. Robustness to white noise (error bars: participant-level 95% CI)



White noise	Clean-trained RF F1 (95% CI)	Noise-augmented RF F1 (95% CI)	Gain (95% CI)
none	0.49 (0.39–0.59)	0.50 (0.40–0.59)	+0.01 (−0.01 to +0.03)
20 dB	0.14 (0.06–0.22)	**0.45 (0.34–0.54)**	+0.30 (0.24–0.38)
10 dB	0.00	**0.39 (0.30–0.49)**	+0.39 (0.30–0.49)
5 dB	0.00	**0.31 (0.21–0.42)**	+0.31 (0.21–0.42)
0 dB	0.00	0.17	–

A detector trained only on clean audio stops working as soon as moderate noise is added. Training with noise keeps it usable down to about 5 dB.

5.3 Does SWALLOHear know when it is wrong? (RQ2)

5.3.1 Raw confidence vs the learned reliability score

Figure 5. Which score tells correct from wrong detections?



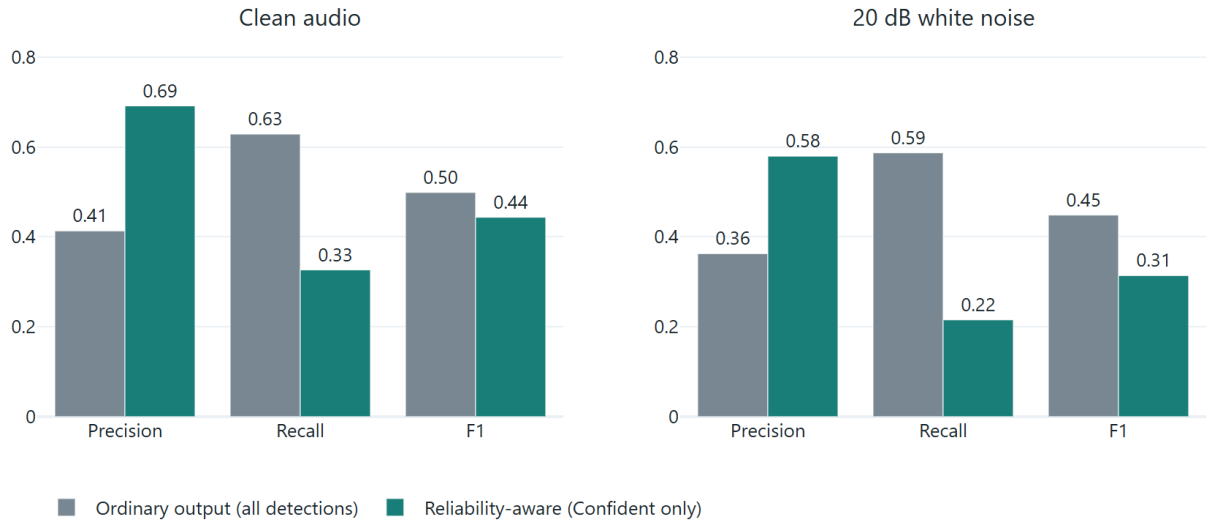
AUROC measures how well a score ranks correct detections above wrong ones (0.5 = coin flip, 1.0 = perfect).

Noise	Raw detector confidence	Learned reliability score	Difference (95% CI)
none	0.53 (0.46–0.60)	**0.79 (0.73–0.83)**	+0.25 (0.17–0.34)
20 dB	0.53 (0.46–0.60)	**0.79 (0.72–0.85)**	+0.26 (0.16–0.37)
10 dB	0.46 (0.38–0.54)	**0.79 (0.72–0.85)**	+0.33 (0.23–0.42)
5 dB	0.41 (0.32–0.48)	**0.78 (0.72–0.83)**	+0.37 (0.28–0.47)

The detector's own confidence is barely better than chance on clean audio and becomes *worse* than chance in noise. The learned score stays at about 0.78–0.79 at every level, and every difference is clearly above zero.

5.3.2 Ordinary vs reliability-aware output

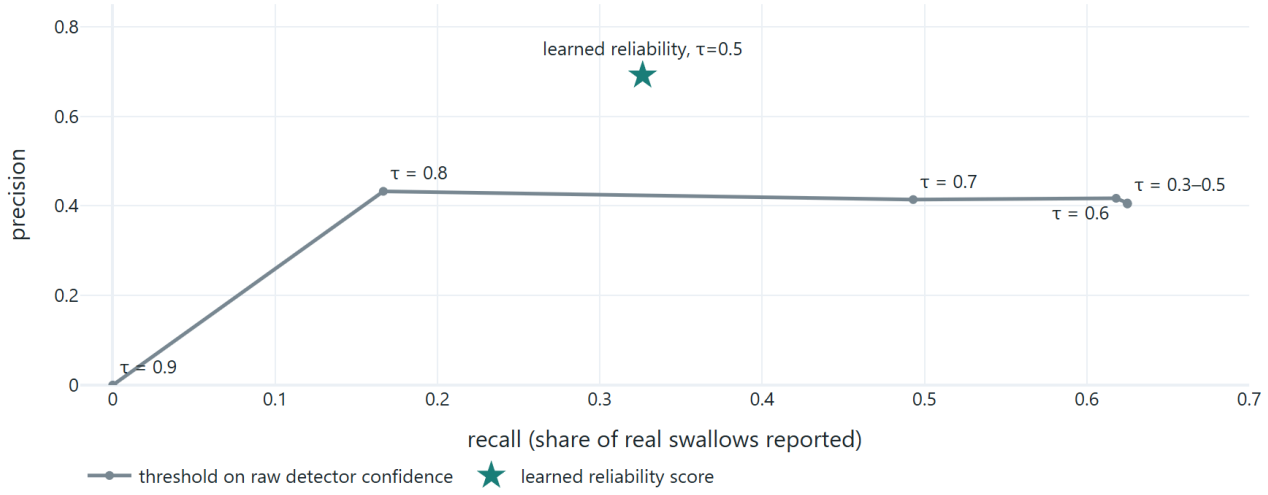
Figure 6. Reporting only Confident events trades recall for precision



Condition	Output	Reported events	Precision (95% CI)	Recall	F1	False detections / min
Clean	Ordinary (all detections)	438	0.41 (0.32–0.51)	0.63	0.50	9.4
Clean	**Reliability-aware** (Confident only)	136	**0.69 (0.59–0.78)**	0.33	0.44	**1.5**
20 dB	Ordinary	466	0.36 (0.26–0.46)	0.59	0.45	10.9
20 dB	**Reliability-aware**	107	**0.58 (0.42–0.70)**	0.22	0.31	**1.6**

Reporting only Confident events raises precision by about 0.28 and removes about 8 false detections per minute (95% CI 6.2–9.8), but finds about half as many real swallows. **This is a trade-off, not a free improvement**: it suits uses where a false alarm is worse than a miss.

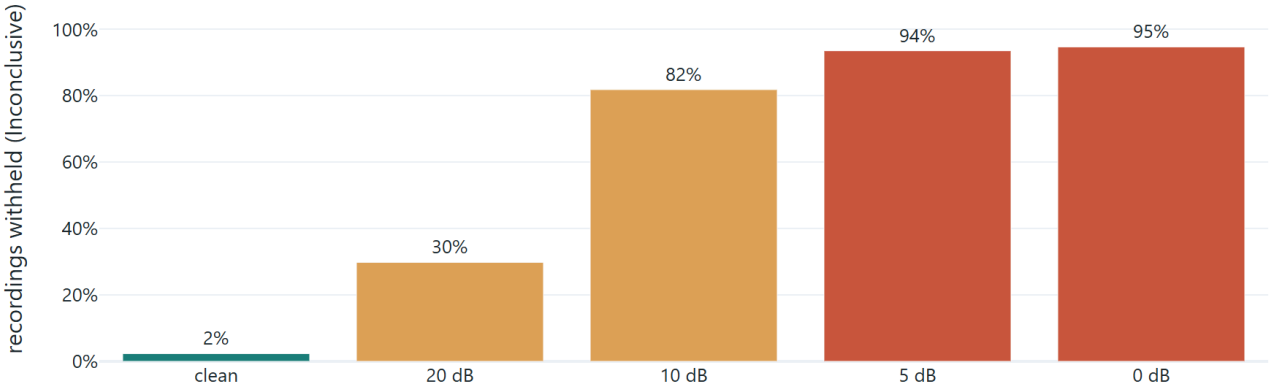
Figure 8. Filtering by raw confidence barely raises precision; the learned score does



Raising a threshold on the detector's own confidence cannot reach this point: its precision stays near 0.41–0.43 whatever the threshold (Figure 8).

5.3.3 The quality gate under noise

Figure 7. Share of test recordings the quality gate withholds



Noise	none	20 dB	10 dB	5 dB	0 dB
Recordings withheld (Inconclusive)	2%	30%	82%	94%	95%

On clean audio the gate almost never interferes. As noise rises, it withholds most recordings, so at 10 dB and below the system mostly refuses to answer instead of reporting results it cannot trust (12 events reported at 10 dB, 3 at 5 dB).

5.3.4 Calibration

Temperature scaling did not fix calibration: the Random Forest's frame-level ECE went from 0.224 to 0.219, and the energy baseline's got worse (0.061 → 0.114). The learned reliability score has an event-level ECE of 0.14 on clean audio and 0.09–0.13 under noise.

5.4 Microphone vs accelerometer

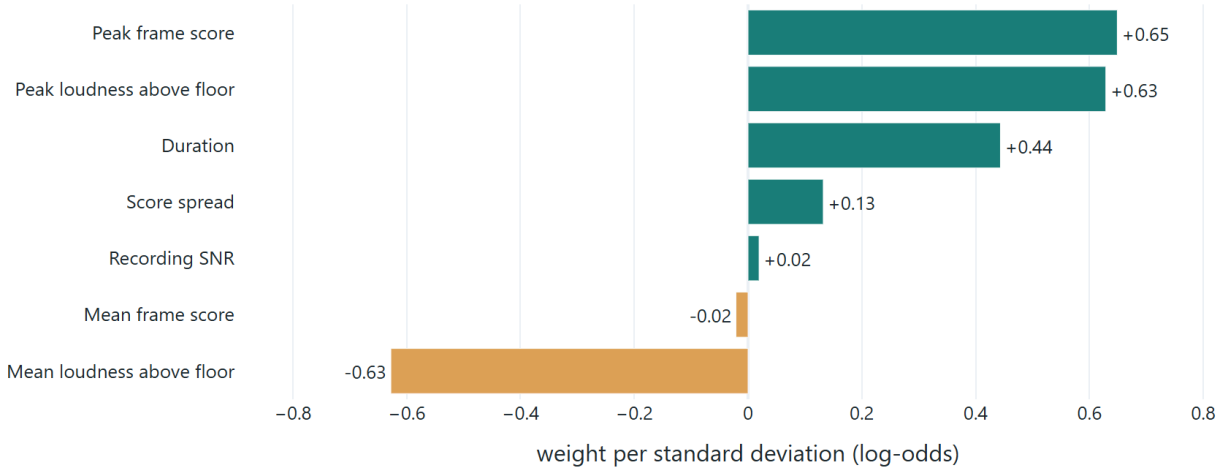
Using the anterior–posterior accelerometer instead of the microphone gave F1 0.47 against 0.49: **no advantage** in this pipeline, which supports microphone-only (phone or stethoscope) research.

5.5 Participant groups

F1 of the final detector was 0.50 for the `ns` group and 0.49 for the `s` group on clean audio. Because the meaning of these groups is not documented, **no stroke-specific conclusion is drawn**.

5.6 What the reliability model learned

Figure 9. What the reliability model has learned



Events are judged more reliable when the detector's **peak** score is high, the event has a clear **loud peak** above the noise floor and it **lasts longer**. A high *average* loudness lowers reliability: broad, uniformly loud stretches tend to be noise or speech rather than a swallow. The detector's mean score (its "confidence") carries almost no weight, which is consistent with section 5.3.1.

5.7 Speed (RQ3)

Analysing a 30-second recording took a **median 0.13 s** (20 runs, maximum 0.16 s) on a laptop CPU (AMD Ryzen 5 6600H), about 236 times faster than real time. The proposal's target was ≤ 5 s.

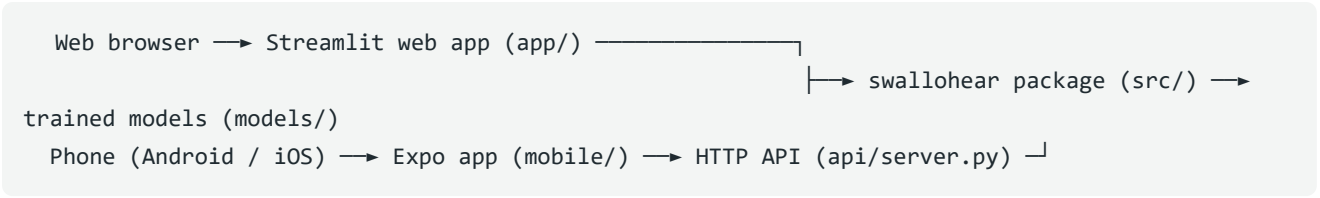
5.8 Hypotheses and objectives

Hypothesis	Outcome
H1 CNN beats the best conventional detector	**Not tested**: the CNN was not built. The Random Forest clearly beat the energy baseline (F1 0.50 vs 0.28; confidence intervals do not overlap).
H2 At 5 dB, Confident events have higher precision than all detections	**Not supported at 5 dB**: the gate withholds 94% of recordings, so only 3 events were reported and precision cannot be estimated. **Supported at clean and 20 dB** (0.41 → 0.69 and 0.36 → 0.58).
H3 Temperature scaling lowers calibration error	**Not supported** (0.224 → 0.219 for the Random Forest; worse for the baseline). A learned reliability model was needed instead.

Objective (proposal)	Status
O1 Data audit (checksums, parse all files, log exclusions)	Done: section 3.3
O2 Detectors with event metrics on a sealed test set	Done for M0 and M1; M2 (CNN) not built
O3 Comparison with participant-level bootstrap CIs and ablations	Done: noise augmentation and sensor channel; filter and context-length ablations not run
O4 Calibration, risk–coverage, noise at 20/10/5/0 dB, quality index	Done (Brier score not reported)
O5 Web application (≥ 6 demo clips, uploads, four-state output, exports, ≤ 5 s latency)	Done (7 clips, 0.13 s), plus a mobile app and API beyond the proposal
O6 Reproducibility (pinned environment, seeds, split, tests)	Done; the split is regenerated deterministically rather than shipped as a file; no single `make` command yet

6. The SWALLOHear system

6.1 Architecture



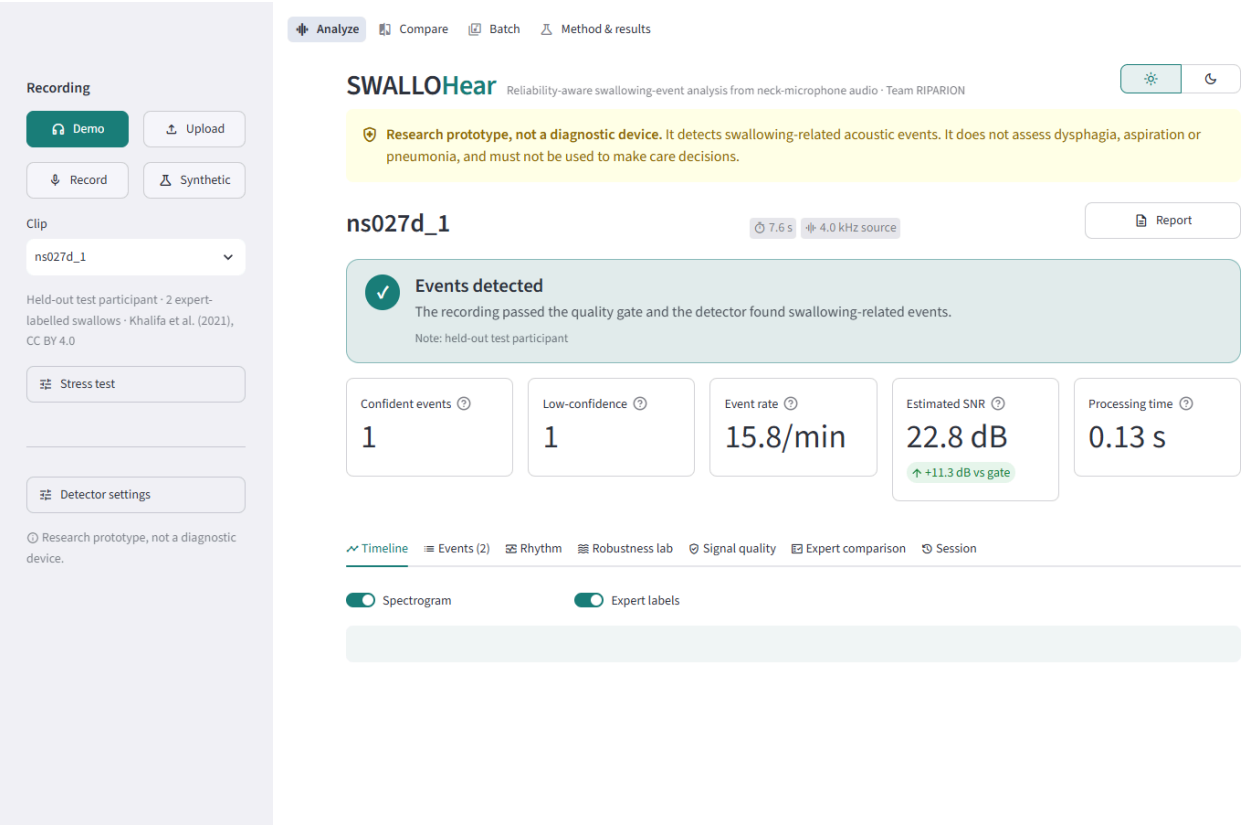
Component	Technology	Role
Core package `src/swallohear/`	Python, NumPy, SciPy, scikit-learn	Loading, preprocessing, features, detectors, quality gate, reliability (`final.py`)
Web app `app/`	Streamlit, Plotly	Four pages, all features, runs the models in-process
HTTP API `api/server.py`	Starlette, uvicorn, imageio-ffmpeg	Serves the same analysis to the mobile app; decodes phone audio
Mobile app `mobile/`	Expo SDK 58, React Native, react-native-svg, expo-audio	Android, iOS and web from one codebase

To keep interaction instant, the analysis is split into a slow step (preprocessing, quality and the Random Forest, cached per recording) and a fast step (gate, events, reliability). Moving a threshold slider re-runs only the fast step.

6.2 The web app

The web app has four pages in a top navigation bar, a sidebar for inputs and settings, and a light/dark switch.

Analyze page



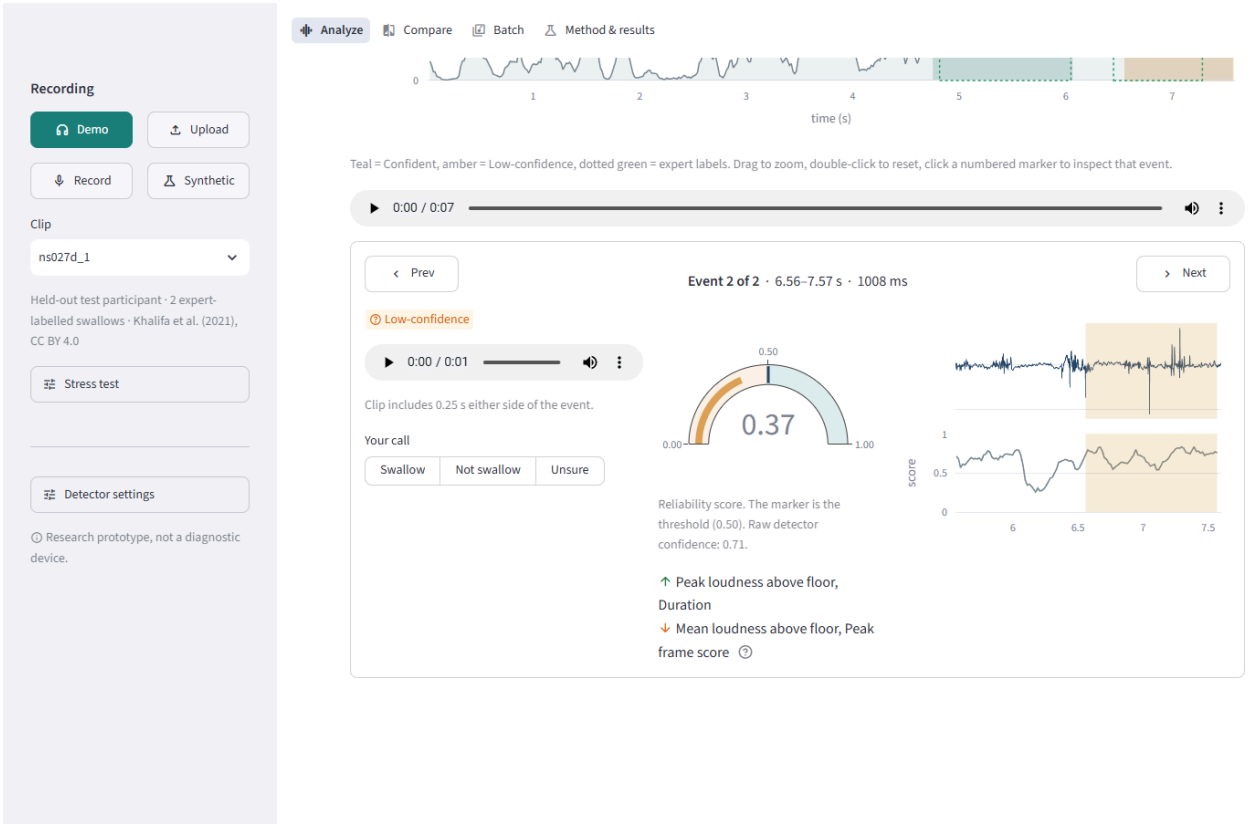
Area	What it does
Recording (sidebar)	Choose **Demo**, **Upload**, **Record** (microphone) or **Synthetic** (generated test bursts).
Stress test (sidebar)	Add white, pink or 60 Hz mains-hum noise at any SNR, or simulate clipping, to see how the system reacts.
Detector settings (sidebar)	Live sliders for the reliability threshold, start and continue thresholds, merge gap and minimum SNR; **Reset to validated**.
Verdict banner	Events detected / No event / Result withheld, with reason codes.
Summary cards	Confident events, low-confidence events, event rate, estimated SNR vs the gate, processing time.
Report button	Downloads the printable report (section 6.5).

The page has seven tabs:

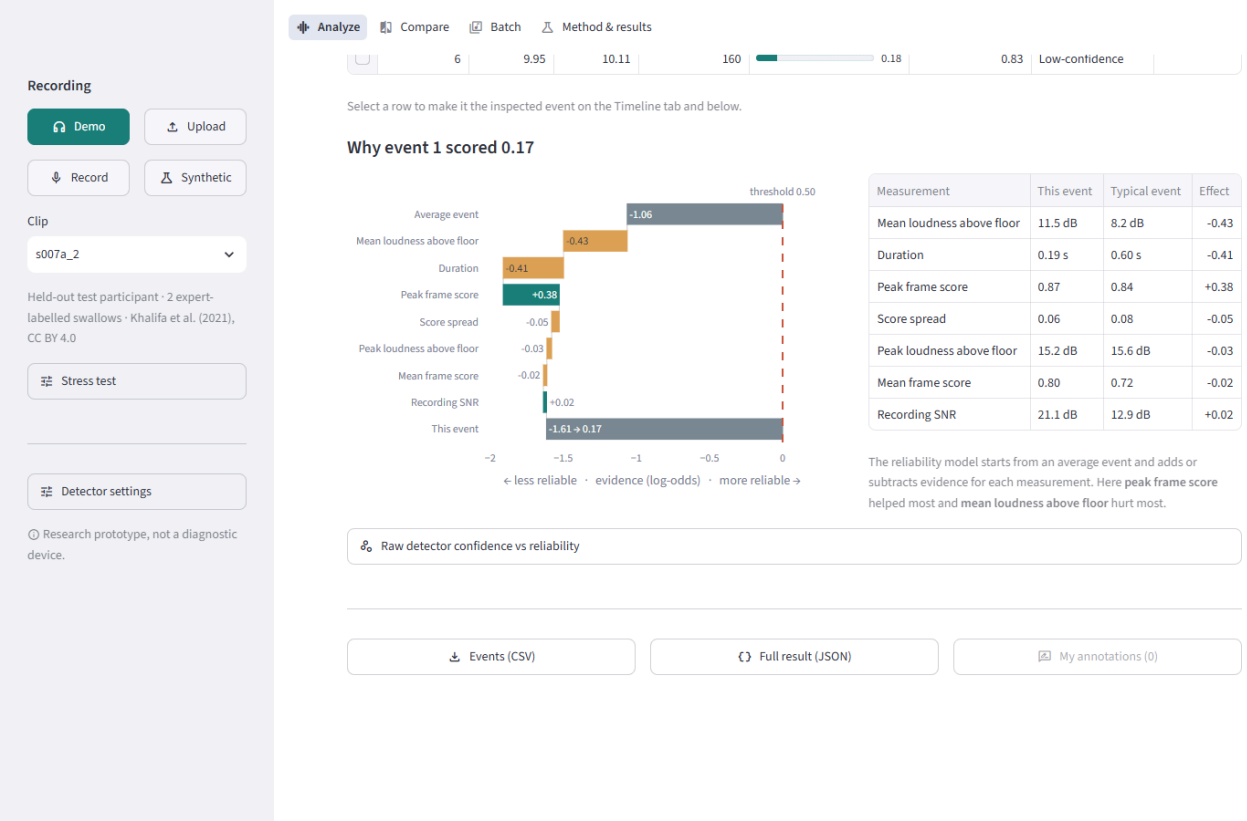
Timeline. Waveform, spectrogram and detector score on one zoomable time axis. Shaded bands are events (teal = Confident, amber = Low-confidence); dotted green outlines are expert labels. Click a numbered marker to inspect an event.



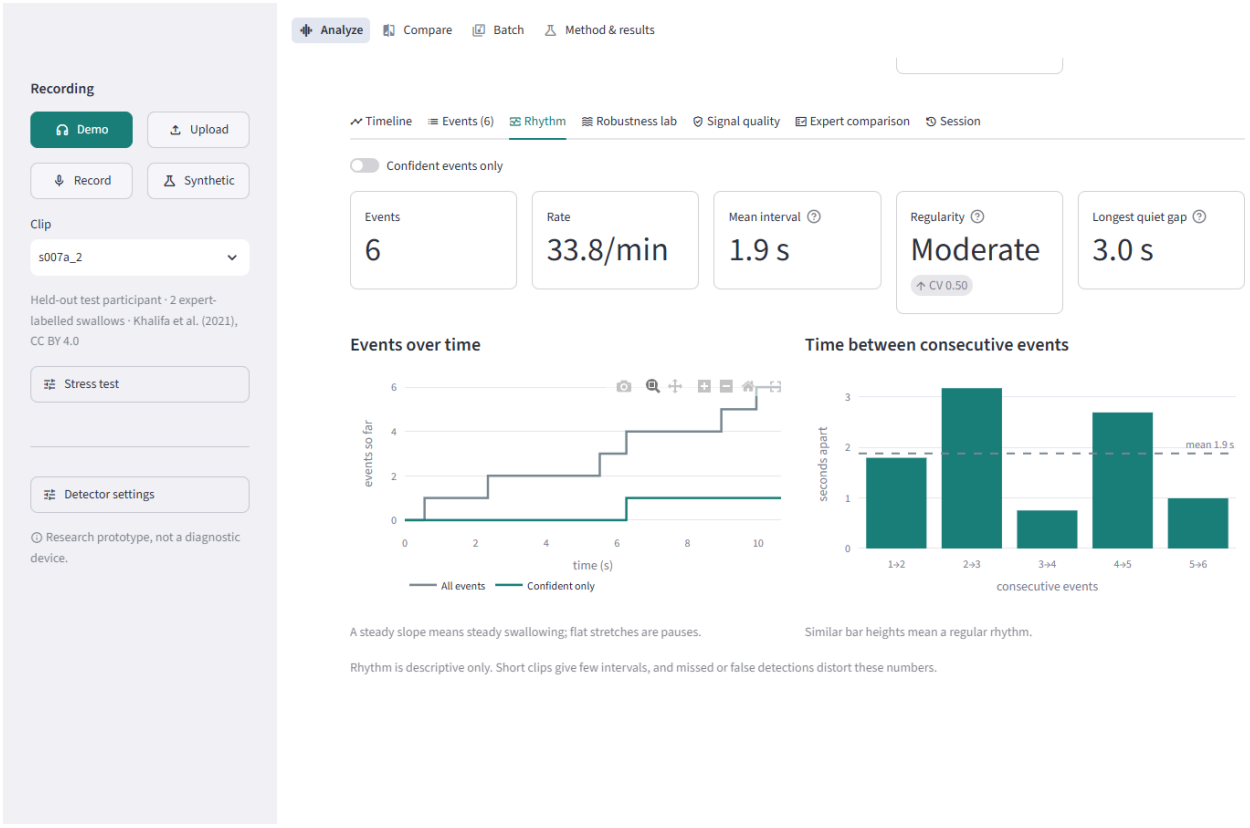
Event card (under the timeline): play just that event, see its reliability gauge against the threshold, the top reasons (↑ raised / ↓ lowered), a zoomed view, and record **your call**: Swallow, Not swallow or Unsure.



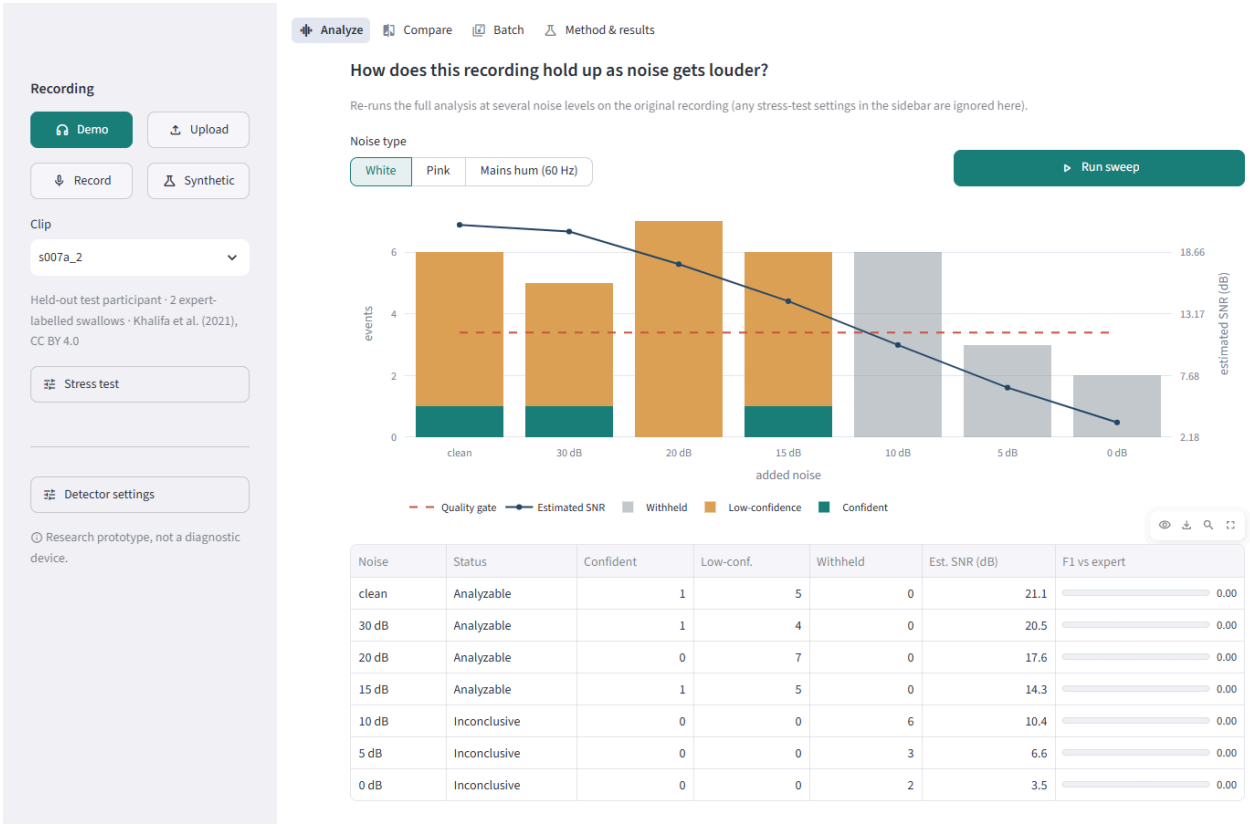
Events. A sortable table of all events; select a row to inspect it. **Why this score?** shows the exact waterfall breakdown (section 4.8) and a table of this event's measurements against a typical event.



Rhythm. Number of events, rate, mean interval, regularity and longest quiet gap, with *events over time* and *time between events* charts and a "Confident only" switch.



Robustness lab. Re-analyses the recording at seven noise levels (clean, 30, 20, 15, 10, 5, 0 dB) and shows how the events change and where the gate starts withholding.



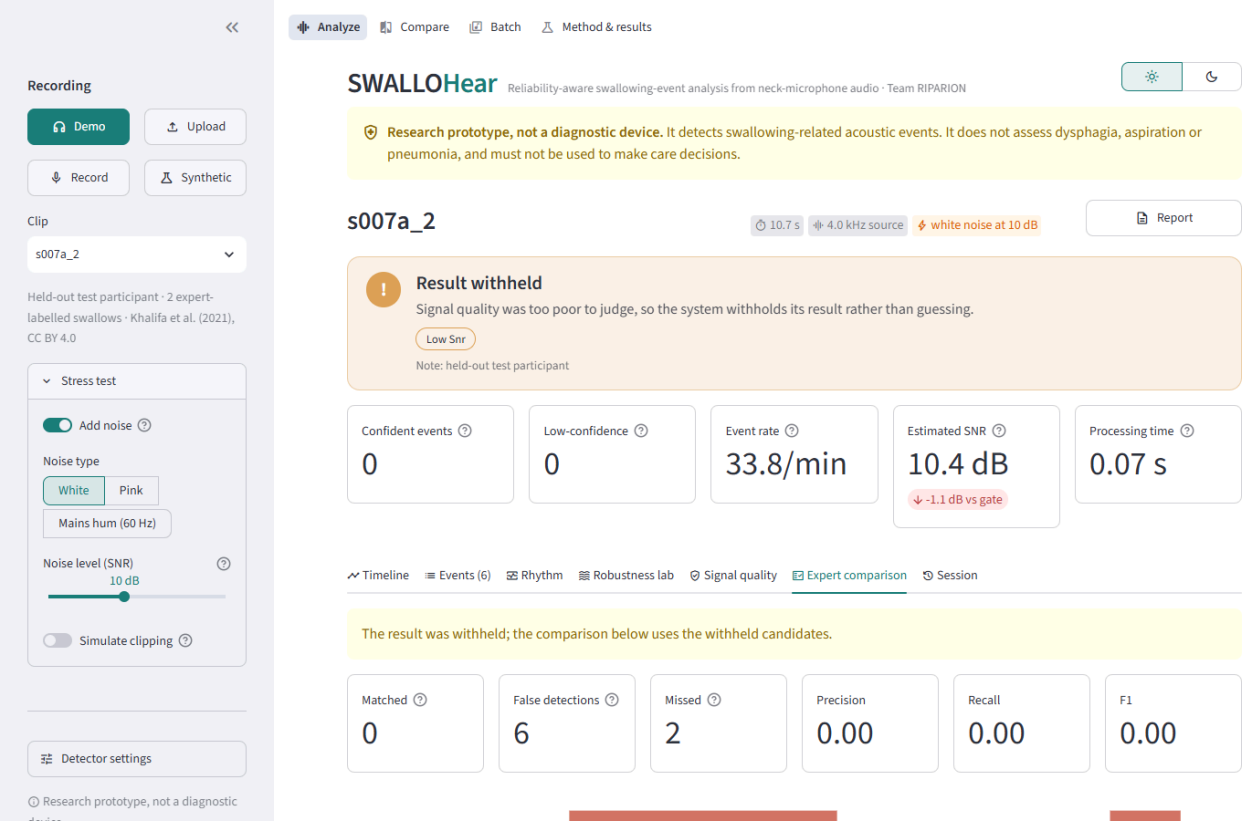
Signal quality. Every gate check with its value and limit, a histogram explaining the SNR estimate, and the average spectrum (mains hum shows as spikes at 60, 120, 180 Hz).

Expert comparison. For labelled clips: matched events, false detections, misses, precision, recall, F1, timing error and *Expert vs System* lanes.

Session. Every analysis run in this session, with exports of the log and all your calls.

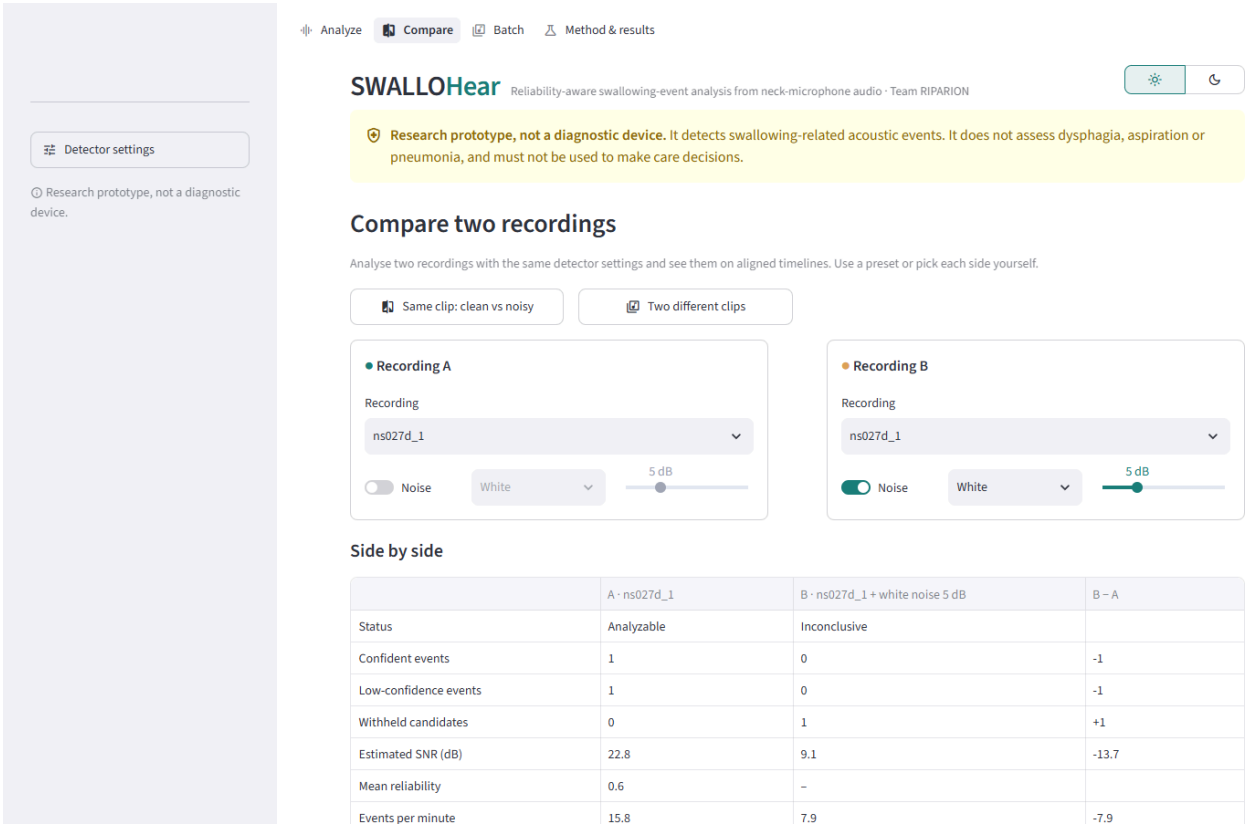
Stress test in action

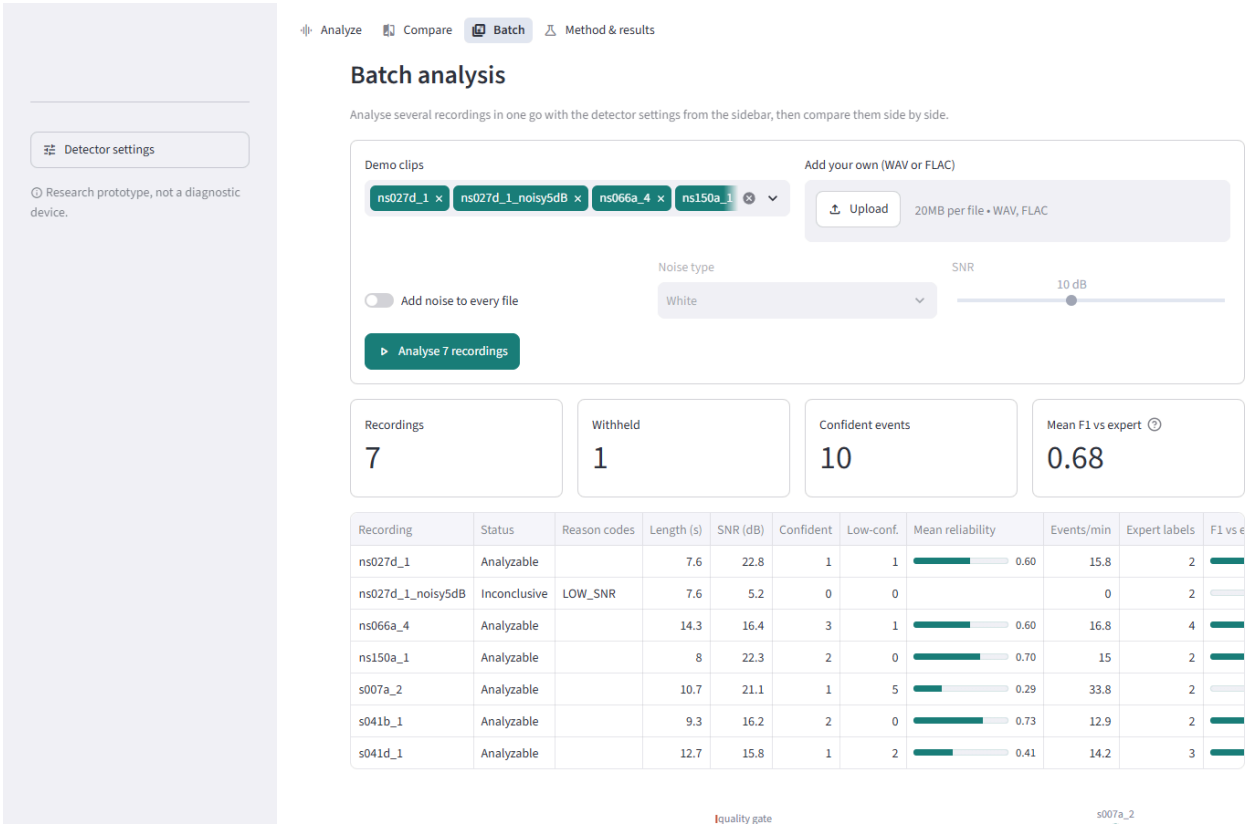
Adding 10 dB white noise to a clean clip drops its estimated SNR below the gate, and the verdict becomes **Result withheld** with the reason *Low SNR*:



Compare, Batch and Method pages

Page	What it does
Compare	Two recordings, or one clip clean vs noisy, side by side: a comparison table with a B − A column, both detector scores overlaid, and aligned timelines.
Batch	Analyse many recordings (demo clips and/or uploads, optional noise) in one go: totals, a sortable table, an SNR-vs-events chart and CSV/JSON export.
Method & results	The pipeline and interactive charts of the experiment results.

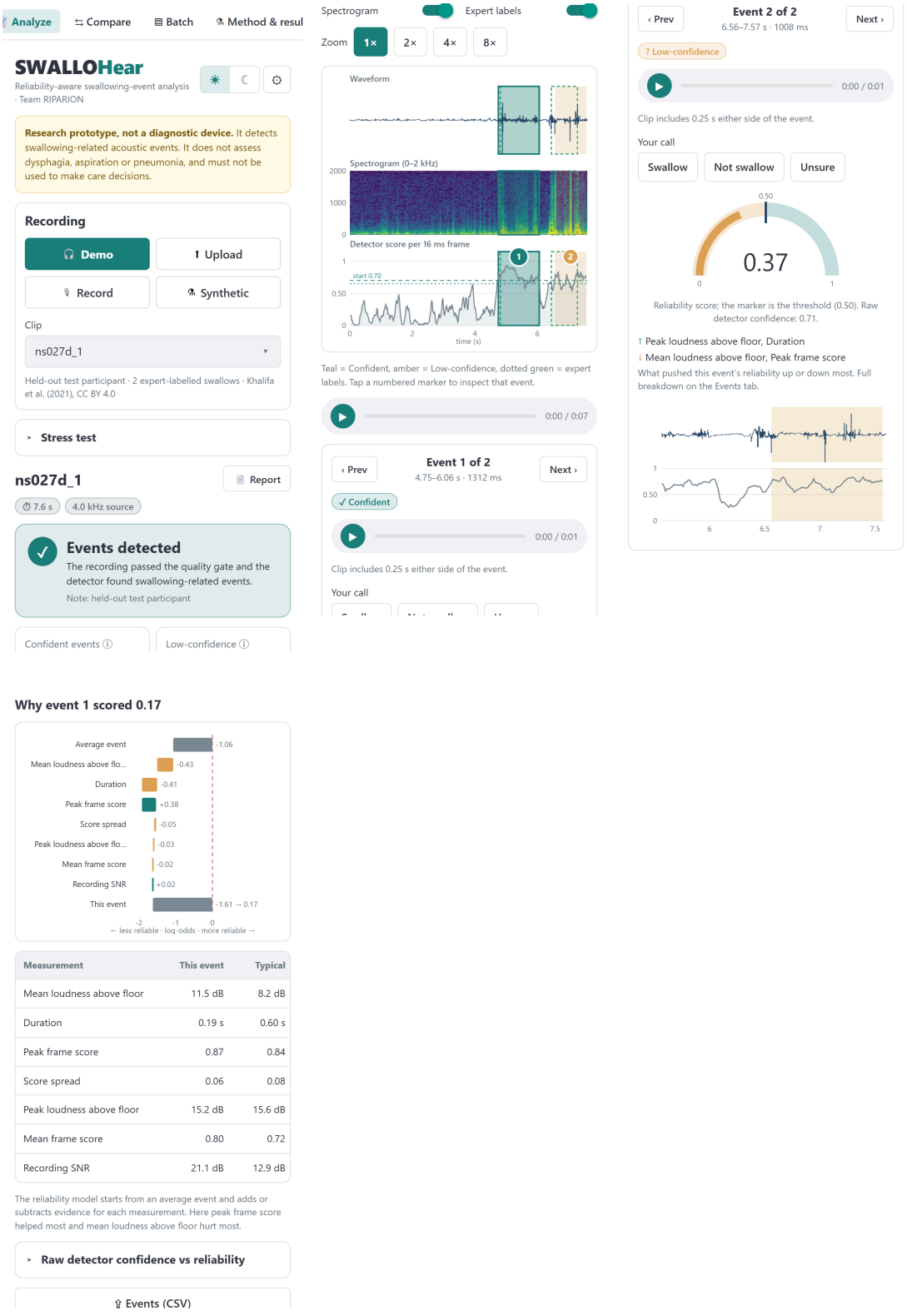


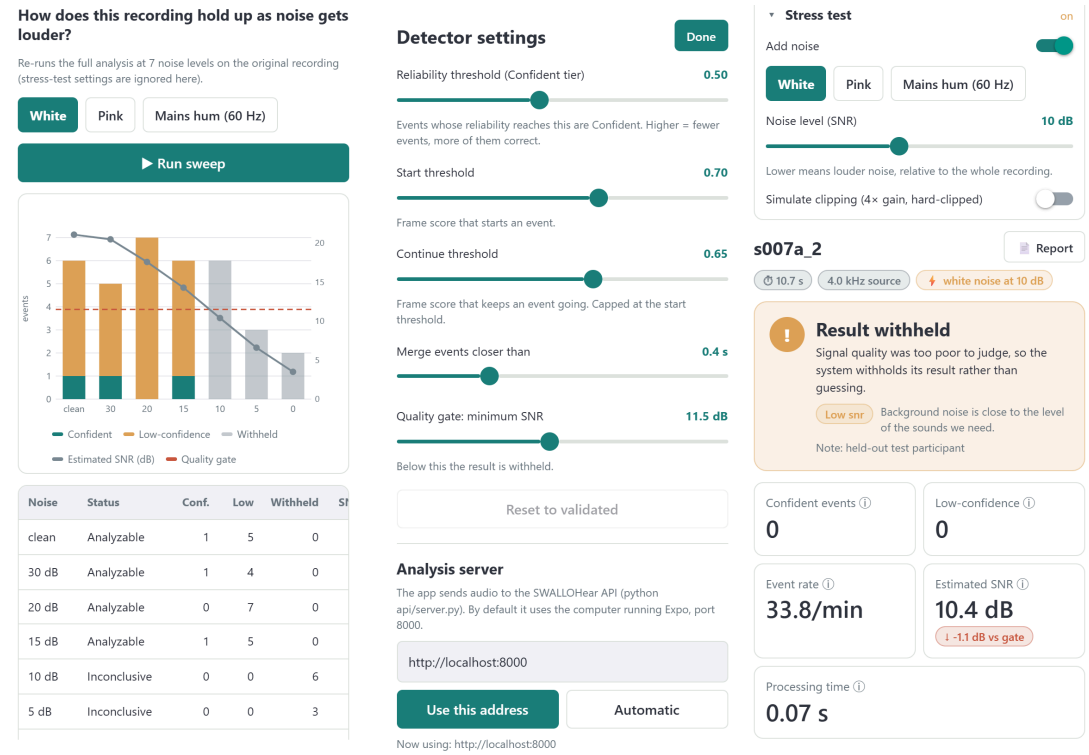


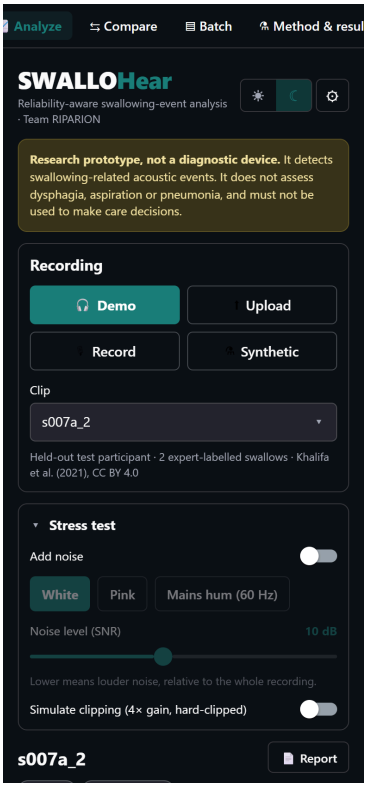
6.3 The mobile app

The mobile app (Android and iPhone, built with Expo) has **every web feature**, in the same colours and layout ideas, adapted for touch: a bottom tab bar (Analyze, Compare, Batch, Method), a ☀/☾ theme switch

and a ⚙ settings sheet in the header, tappable event markers with 1–8× zoom, and the phone's share sheet for exports. It can record directly from the phone's microphone.







The phone sends the audio to the API on a computer on the same Wi-Fi, which runs the same Python models, so results are identical to the web app. The app finds the server automatically when opened through Expo Go.

6.4 The HTTP API

Endpoint	Purpose
`GET /api/health`	Is the server running?
`GET /api/meta`	Validated settings, demo clips, feature names, experiment results
`POST /api/load`	Load a demo clip, synthetic signal or uploaded file (any common audio format); returns a recording id
`POST /api/analyze`	Analyse a loaded recording with given settings and stress test; returns everything the screens show
`POST /api/sweep`	Robustness lab: analyse at seven noise levels
`POST /api/batch`	Batch analysis
`GET /api/analysis/{id}/audio.wav`	The analysed audio, or one event's clip (`?t0=&t1=`)
`POST /api/analysis/{id}/notes` , `GET …/report.html`	Save your calls and open the printable report

6.5 Exports and the report

Export	Contents
Events (CSV)	One row per event: times, duration, reliability, raw confidence, tier, your call
Full result (JSON)	Status, reasons, settings, stress test, events, withheld candidates, quality, annotations
My annotations / All annotations (CSV)	Only the events you gave a call
Session log (CSV)	Every analysis in the session
Batch results (CSV / JSON)	One row per recording
Report (HTML → PDF)	Disclaimer, verdict, summary, timeline, events over time, event table with reasons and your calls, rhythm, expert agreement, quality gate, settings and limitations. Works offline; open it and print to PDF.

6.6 Privacy and security

Audio is processed in memory and is not written to disk or logged; the API keeps only the last 40 recordings in memory and forgets them on restart. File type, size (≤ 20 MB) and length (≤ 120 s) are checked. There are no accounts. The apps warn users not to upload identifiable patient audio to a shared server.

7. User guide

*Step-by-step instructions for anyone, with no programming experience needed. Commands are typed into a terminal (Windows: **Command Prompt**), one at a time, pressing Enter after each.*

7.1 What you need

For	You need
Web app	A Windows, macOS or Linux computer; **Python 3.13**; about 1 GB free space
Mobile app	Also **Node.js 22 or 24 LTS** on the computer, and the free **Expo Go** app on the phone (App Store / Google Play)

On Windows, install Python 3.13 from python.org or type `py install 3.13`. Install Node.js from nodejs.org (choose "LTS").

7.2 One-time installation

1. Download the project: on https://github.com/jedelacruz/swallohear click **Code** → **Download ZIP** and unzip it, or run `git clone https://github.com/jedelacruz/swallohear.git`.
2. Open a terminal **in the project folder** (in File Explorer, open the folder, type `cmd` in the address bar, press Enter).
3. Create a private Python environment for the project: `py -3.13 -m venv .venv`
4. Switch it on (do this every time you open a new terminal): `.venv\Scripts\activate` The line now starts with `(.venv)`. On macOS/Linux: `source .venv/bin/activate`.
5. Install everything SWALLOHear needs (takes a few minutes): `pip install -r requirements.txt`
6. Optional check (should say "8 passed"): `python -m pytest -q`

7.3 Start the web app

```
streamlit run app/streamlit_app.py
```

Your browser opens **http://localhost:8501**. To stop, click the terminal and press **Ctrl+C**. Other devices on the same Wi-Fi can open the *Network URL* printed in the terminal.

7.4 Start the mobile app

You need **two** terminals, both in the project folder with `.venv\Scripts\activate` run first.

Terminal 1, the analysis server:

```
python api/server.py
```

Wait for `SWALLOHear API on http://0.0.0.0:8000`. If Windows asks whether Python may use the network, tick **Private networks** and click **Allow**.

Terminal 2, the app:

```
cd mobile
npm install
npx expo start
```

(`npm install` only the first time.) A QR code appears.

- **Android:** open Expo Go and tap **Scan QR code**.
- **iPhone:** first create a free account at expo.dev/signup and sign in to it in Expo Go; on the computer run `npx expo login --no-browser` with the same account; then scan the QR code with the **Camera** app.
- **No phone:** press **w** in Terminal 2 to open the mobile app in your browser.

The phone and the computer must be on the **same Wi-Fi**. You can use the web app, the phone and the browser version at the same time.

7.5 Your first analysis (five minutes)

1. Open the app. **Demo** is selected and the first clip (`ns027d_1`) is analysed automatically.
2. Read the **verdict**: "Events detected".
3. Read the **cards**: 1 Confident event, 1 Low-confidence event, estimated SNR 22.8 dB (11.3 dB above the gate).
4. In the **Timeline** tab, click (or tap) the numbered marker **2**. The event card shows a reliability of **0.37**: below the 0.50 threshold, so it is *Low-confidence*.
5. Press ▶ in the event card to **listen** to it.
6. Under **Your call**, choose *Swallow*, *Not swallow* or *Unsure*.
7. Open the **Events** tab and look at **Why event 2 scored 0.37** to see which measurements pulled it down.
8. Open **Expert comparison**: both events match the expert labels (F1 1.00 for this clip).
9. Open **Stress test**, switch on **Add noise** at 10 dB: the verdict changes to **Result withheld** (*Low SNR*). Switch it off again.

7.6 Reading the results

You see	It means
Teal band / ✓ Confident	Reliability ≥ 0.50: the system stands behind this detection (about 7 in 10 are correct on clean test audio).
Amber band / ? Low-confidence	Detected, but not trusted: check it yourself.
Faded bands + "Result withheld"	Quality was too poor; the bands are shown only for inspection.
Dotted green outline	Where an expert marked a swallow (demo clips only).
Detector score line	How swallow-like each 16 ms moment sounds (0 to 1); dashed lines are the start/continue thresholds.
Estimated SNR card	Green ↑ = above the quality gate; red ↓ = below it (result withheld).
↑ / ↓ reasons	The measurements that raised or lowered an event's reliability most.

7.7 Common tasks

Analyse your own recording. Choose **Upload** and pick a WAV or FLAC file (web) or any common audio file (mobile), up to 120 s and 20 MB.

Record with the microphone. Choose **Record**. Rest the microphone lightly against the side of the throat, below the jaw, in a quiet room; stay still and quiet; press record and swallow a few times (e.g. sips of water); then stop. The recording is analysed straight away. A phone's built-in microphone is *not* a validated sensor: treat results as a demonstration.

Test how robust it is. Use **Stress test** (noise or clipping) for one setting, or the **Robustness lab** tab → **Run sweep** for seven noise levels at once.

Change the detector settings. Open **Detector settings** (web sidebar, or ⚙ on mobile). A higher *reliability threshold* means fewer but more trustworthy Confident events; higher *start/continue thresholds* mean fewer, shorter events. **Reset to validated** restores the values the results were measured with.

Compare two recordings. Open **Compare**, pick A and B (or press *Same clip: clean vs noisy*), and read the B − A column.

Analyse many recordings. Open **Batch**, select clips and/or add files, press **Analyse**, then export CSV or JSON.

Review and annotate. Step through events with **Prev / Next** and set *Your call* on each; your calls appear in the events table, the exports and the report.

Export and print. Events tab → **Events (CSV)**, **Full result (JSON)**, **My annotations**. **Report** (next to the recording name) → opens the report → **Ctrl+P** → **Save as PDF**. On a phone, exports open the share sheet.

Dark mode. Click ☾ in the top-right corner (☀ for light).

7.8 Frequently asked questions

Is this a diagnosis? No. It detects sounds and reports how far to trust them. It cannot tell whether a person has dysphagia or is at risk of aspiration.

Why did it say "Result withheld"? The recording failed a quality check, usually *Low SNR* (too much background noise). Record again in a quieter place with the microphone in firm, gentle contact with the skin.

Why are some events Low-confidence? The reliability model judged them unlikely to be correctly placed swallows, for example because they are short, faint or uniformly loud. They may still be real: listen and decide.

Why do the numbers differ from the paper when I move sliders? The paper uses the validated settings. Press **Reset to validated** to return to them.

Is my audio saved? No. It is analysed in memory and forgotten. Your calls and the session log stay only in the open app; export them to keep them.

Does it work offline? The web app runs entirely on your computer. The mobile app needs the computer running the API on the same Wi-Fi.

7.9 Troubleshooting

Problem	Fix
`DLL load failed … Application Control policy has blocked this file`	Windows Smart App Control blocked a very new package. Use Python 3.13 and `pip install -r requirements.txt`, which pins versions that load.
`'streamlit'` or `'python'` is not recognized	Open the terminal in the project folder and run `.venv\Scripts\activate`.
`'Analyzer' object has no attribute 'score'`	The server was started before the code was updated: press Ctrl+C and start it again.
Phone: "Can't reach the analysis server"	Make sure `python api/server.py` is running, both devices are on the same Wi-Fi, and Python is allowed through the Windows firewall (*Private networks*). Or set the address under ⚙ → Analysis server.
iPhone: "You need to be signed in to Expo Go and Expo CLI"	Sign in to the same Expo account in Expo Go and with `npx expo login --no-browser`.
Phone cannot open the QR code on school or public Wi-Fi	Use a phone hotspot for both devices, or `npx expo start --tunnel` and set the server address manually.
Blank page after restarting	Reload the browser tab.

8. Discussion

8.1 What the results mean

Detection works, modestly. A Random Forest on hand-crafted features finds about 63% of labelled swallows on unseen participants, with 4 in 10 reported events correct. This is well above an energy threshold, but far from clinical standard.

Noise is the main threat, and training with it is cheap insurance. The clean-trained detector failed completely at 10 dB; adding noisy copies during training kept it working, with no cost on clean audio.

The detector does not know when it is wrong; a second model does. This is the central finding. The detector's confidence carries almost no information about correctness (AUROC 0.53) and calibration could not repair it. A small, transparent second model using simple properties of each event (peak score, peak loudness, duration) separated correct from wrong detections well (AUROC 0.79) and kept doing so in noise. Combined with the quality gate, the system turns "silent failure" into either a flagged low-confidence event or an explicit refusal.

Reliability has a price. Reporting only Confident events nearly doubles precision but halves recall. Which output to use depends on the application, and the apps show both tiers so the user can decide.

8.2 Limitations

1. **Modest absolute accuracy** (event F1 ≈ 0.5).
2. **One dataset, one split, 27 test participants**: confidence intervals are wide, and results may not transfer to other populations, rooms or sensors.
3. **No clinical or Philippine validation**; no phones or low-cost stethoscopes were tested against reference sensors.
4. **Synthetic noise** (added white noise) only, not real wards or homes.
5. **Possibly incomplete labels**: the dataset authors report unlabelled swallows, so some "false detections" may be real. The planned manual audit was not done.
6. **No CNN** (M2) and no ensemble-based uncertainty; the filter and context-length ablations were not run.
7. **Unverified groups**: the `ns` / `s` prefixes are not stroke labels, so no stroke-specific claim is made.
8. **Rhythm measures are descriptive** and inherit every missed or false detection.

8.3 Ethics, privacy and safety

Only public, de-identified data were used, under CC BY 4.0 with attribution, and no re-identification was attempted. SWALLOHear is research-only: outputs are for researchers and qualified clinicians and are not a basis for patient management. Uncertainty is shown by design, a permanent banner states the research-only status, and poor recordings are never given a definite result. Audio is processed in memory and not stored. Any future recordings by the team must come from consenting adult volunteers, be non-clinical and be handled under the Philippine Data Privacy Act of 2012.

8.4 Relevance to the Philippines

Instrumental swallowing assessment is concentrated in large hospitals. A transparent, open and low-cost analysis foundation, which runs on an ordinary laptop and a phone, and which says when it cannot be trusted, is a starting point for locally validated research in rehabilitation and neurology units, provided it is first tested on Filipino patients, local devices and real ward noise.

9. Future work

1. Ethics-approved Philippine pilot recordings with partner rehabilitation or neurology units.
2. Clinician-rated reference labels (videofluoroscopy or endoscopy) to study severity, which this work cannot do.
3. Prospective comparison of stethoscopes, smartphones and research sensors.
4. The compact CNN (M2), ensembles for uncertainty, and the remaining ablations.
5. A manual audit of "false detections" to estimate label incompleteness.
6. Offline, on-phone inference for low-connectivity settings, and a Filipino-language interface.
7. Usability testing with speech-language pathologists.
8. Regulatory review before any clinical use.

10. Conclusion

SWALLOHear detects swallowing-related sounds in neck-microphone audio with modest accuracy on unseen participants, and, more importantly, knows when not to be trusted: a quality gate refuses recordings it cannot read, and a learned reliability model separates trustworthy detections from doubtful ones far better than the detector's own confidence, with an exact explanation for every score. The whole system is open, reproducible and usable on an ordinary laptop, in a browser and on a phone, as a foundation for future, locally validated swallowing research.

Acknowledgements

We thank Khalifa, Donohue, Coyle and Sejdić for releasing the dataset under an open licence, and the organisers of the Philippine Junior Data Science Challenge 2026.

Team RIPARION

Jan Emmanuel Dela Cruz · Denver Reyes · Joshua Abalo · Rhaine Cyrel Valera · Jhayciel Santiago

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Appendices

Appendix A. Output states

State	Rule	Shown to the user
Confident event	Quality gate passed; reliability ≥ τ (0.50)	Event with timing, duration and reasons (teal)
Low-confidence event	Quality gate passed; reliability < τ	"Detected, not trusted: review" (amber)
No event detected	Quality gate passed; no event assembled	"Nothing crossed the threshold; does not prove no swallow occurred"
Inconclusive (Result withheld)	Any quality check failed	Reason codes; candidates shown faded for inspection only; re-record advised

Appendix B. Detector settings

Setting	Validated value	Range in the apps	Effect of raising it
Reliability threshold τ	0.50	0.20–0.90	Fewer Confident events, more of them correct
Start threshold	0.70	0.30–0.95	Fewer events start
Continue threshold	0.65	0.20–0.95 (≤ start)	Events end sooner
Merge gap	0.4 s	0–1.5 s	More nearby events joined into one
Minimum SNR (quality gate)	11.5 dB	0–25 dB	More recordings withheld

Appendix C. Reason codes

Code	Meaning
`LOW_SNR`	Background noise is close to the level of the sounds being measured.
`CLIPPING`	The recording is distorted because the input level was too high.
`FLATLINE`	Long stretches of the signal are perfectly flat, usually a dropout.
`LOW_BANDWIDTH`	The file was recorded at too low a sample rate to carry swallow sounds.
`TOO_SHORT` / `TOO_LONG`	Under 1 second / over 120 seconds.

Appendix D. Reliability model weights

Intercept −1.063 (the log-odds of an average event, reliability 0.26).

Measurement	Weight (per SD)	Training average	Training SD
Peak frame score	+0.649	0.840	0.053
Peak loudness above floor	+0.629	15.65 dB	7.84 dB
Duration	+0.444	0.60 s	0.44 s
Score spread	+0.132	0.084	0.052
Recording SNR	+0.019	12.94 dB	7.54 dB
Mean frame score	−0.021	0.720	0.069
Mean loudness above floor	−0.628	8.19 dB	4.77 dB

Appendix E. Repository map

Path	Contents
`src/swallohear/`	`io.py` audio loading · `preprocess.py` filtering, normalisation, framing · `features.py` 81 frame features · `detectors.py` M0, M1 · `postprocess.py` events · `reliability.py` quality indicators · `metrics.py` matching, F1, ECE · `final.py` the analysis used by the apps · `synth.py` synthetic signals · `data.py` dataset access and split
`scripts/`	`prepare_data.py`, `build_index.py`, `evaluate_baselines.py`, `run_experiments.py`, `run_reliability_v2.py`, `bootstrap_ci.py`, `make_demo_clips.py`, `make_paper.py`, `capture_docs.py`, `capture_mobile.py`
`app/`	Streamlit web app (`streamlit_app.py`, `views/`)
`api/server.py`	HTTP API for the mobile app
`mobile/`	Expo / React Native app (`src/app` routes, `src/screens`, `src/components`, `src/lib`)
`models/`, `configs/final.json`	Trained models and validated thresholds
`results/`	`baselines.json`, `experiments.json`, `reliability_v2.json`, `bootstrap.json`
`data/demo/`	Seven demo clips with expert labels
`tests/`	`test_core.py` (8 unit tests incl. split disjointness), `test_api.py` (every API endpoint)
`docs/`	This paper, web and mobile guides, figures, screenshots, demo GIF and video

Appendix F. Reproducing every number

1. Install as in section 7.2.
2. Download `signals.zip` (1.4 GB) and `labels.zip` from DOI 10.5281/zenodo.4539695 into `data/raw/`.
3. Run, in order:

```
 python scripts/prepare_data.py --dir data/raw
python scripts/build_index.py
python scripts/evaluate_baselines.py
python scripts/run_experiments.py
python scripts/run_reliability_v2.py
python scripts/bootstrap_ci.py
python scripts/make_demo_clips.py
python scripts/make_paper.py figures
```

Seeds are fixed; steps 3–6 take about 30 minutes on a laptop CPU. `bootstrap_ci.py` prints "OK" for each noise level when it reproduces the main run. Software versions are pinned in `requirements.txt` (Python 3.13, NumPy 2.2.6, SciPy 1.15.3, scikit-learn 1.6.1, pandas 2.2.3) and `mobile/package-lock.json` (Expo SDK 58).

Appendix G. Glossary

Term	Meaning
AUROC	Area under the ROC curve: the chance a randomly chosen correct detection scores higher than a randomly chosen wrong one.
Bootstrap CI	A confidence interval made by re-sampling test participants many times.
Calibration / ECE	Whether a score of 0.7 is right 70% of the time; expected calibration error measures the gap.
Cervical auscultation	Listening to swallowing at the neck.
dB	Decibel, a logarithmic unit of loudness ratio.
Dysphagia	Difficulty swallowing.
Hysteresis	Using a higher threshold to start an event than to keep it going.
IoU	Intersection over union: overlap of two time intervals divided by their combined span.
MFCC	Mel-frequency cepstral coefficients, a standard description of sound timbre.
Participant-disjoint	No person appears in more than one of training, validation and test.
Random Forest	Many decision trees voting together (Breiman, 2001).
Videofluoroscopy	X-ray video of swallowing; the source of the expert labels.